

A WebXR Client for Exploring Industrial Physical Simulations using Augmented Reality and Hand Gestures

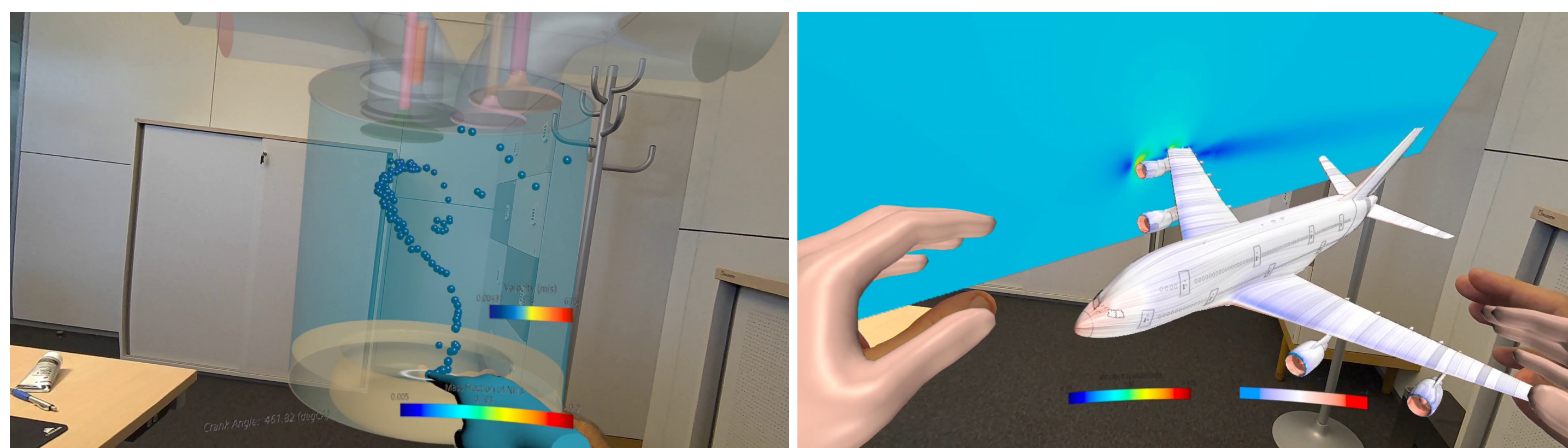
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Motivation

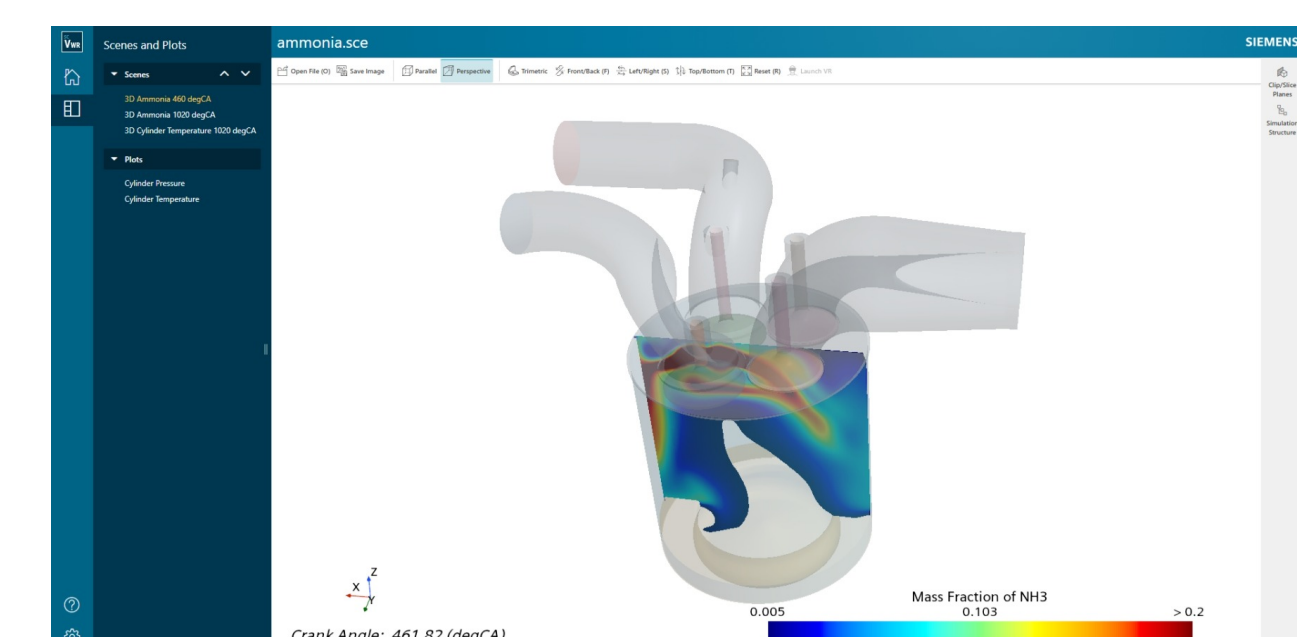
Our objective is to explore complex fluid dynamics simulations as created by Simcenter STAR-CCM+ [1]. While traditional Virtual Reality (VR) is a great exploration tool, we learned that potential customers can suffer from simulation sickness due to being fully immersed in the VR environment [2], and from confusion due to complex controller schemes. Furthermore, Augmented Reality (AR), compared to VR, allows users to view and interact with CFD simulation results in the context of the real-world environment, which can be particularly useful for applications where the physical surroundings are relevant, such as in industrial settings. Recent AR devices with their pass-through video and hand tracking capabilities offer compelling solutions to these problems, and we want to take advantage of this [3].

Our System



We propose a WebXR-based client for interacting with industrial simulations using AR that extends our cloud solution. With this approach a native implementation for a particular mobile device is not required, and thus can make use of devices that support pass-through video and hand tracking like the Meta Quest 3. A main focus was on easy-to-learn hand gestures that naturally fit with the chosen action. Our official web viewer currently supports a VR mode, and will be extended to AR in a future release:

<https://cloud.sw.siemens.com/starccmviewer>



Our simulations typically contain surface geometry on which scalar and vector fields can be displayed, as well as volumetric scalar and vector fields. This data can be explored with the following visualization techniques:

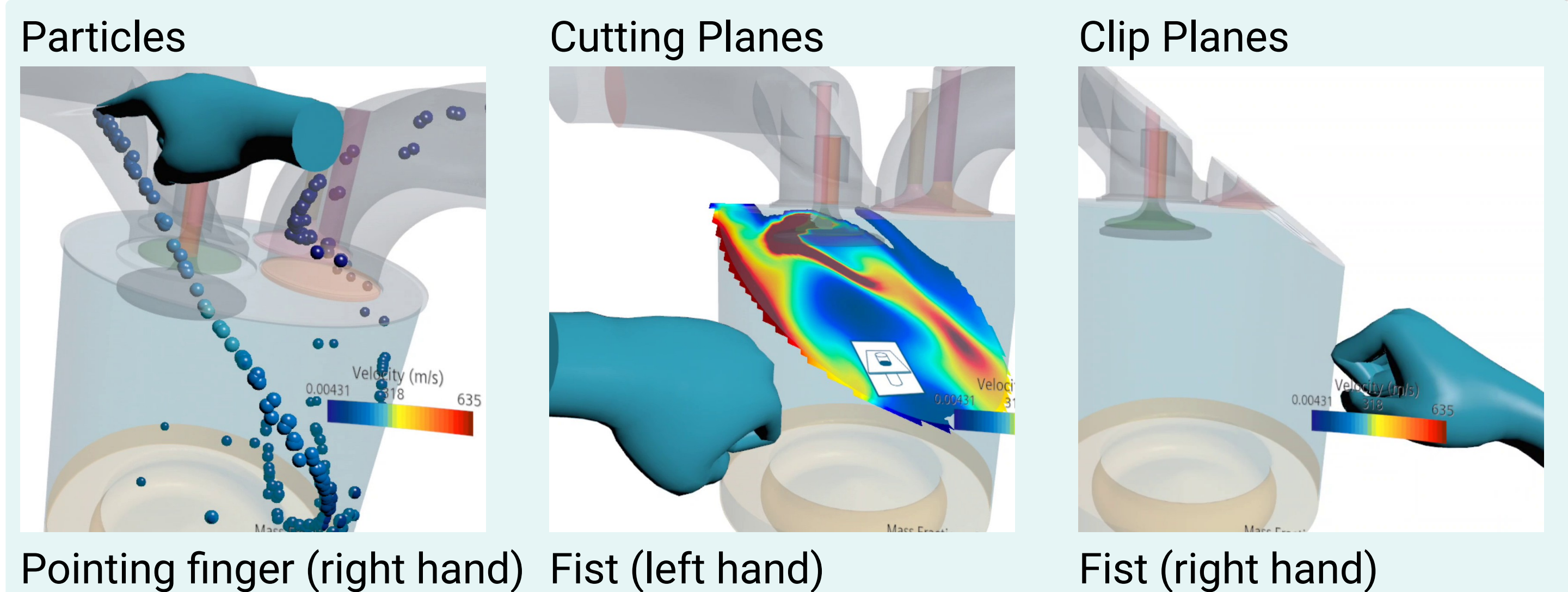
Visualization Techniques

Particles	injected into a vector field, they follow it with a speed proportional to the magnitude of the vector quantity
Cutting Planes	visualize scalar quantities on plane intersecting a scalar volume
Clip Planes	clip away part of geometry to visualize hidden structures

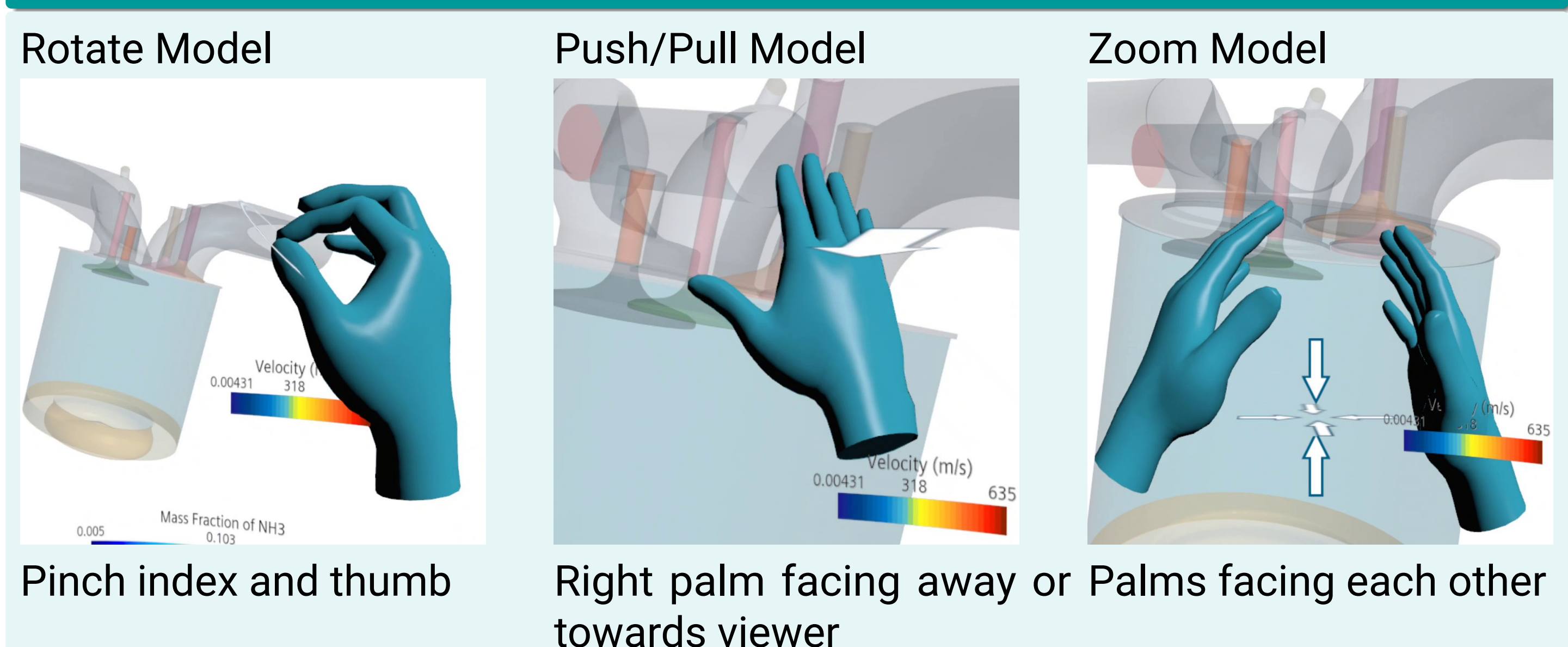
Hand Gestures

We introduce a number of simple and intuitive hand gestures to interact with a model and trigger interactive visualization techniques, in order to give customers and engineers, many of which are first-time AR users, a pleasant experience. For example, particles are injected by pointing into the volume with the index finger, and planes align with the wrist joint of the fist. Particle sources and planes can be made permanent by simply giving a thumbs-up with the other hand. Intuitively, zoom-ins and zoom-outs are controlled by the distance between the palms. The option menu is operated by touching the palm of the left hand with the index finger of the other hand.

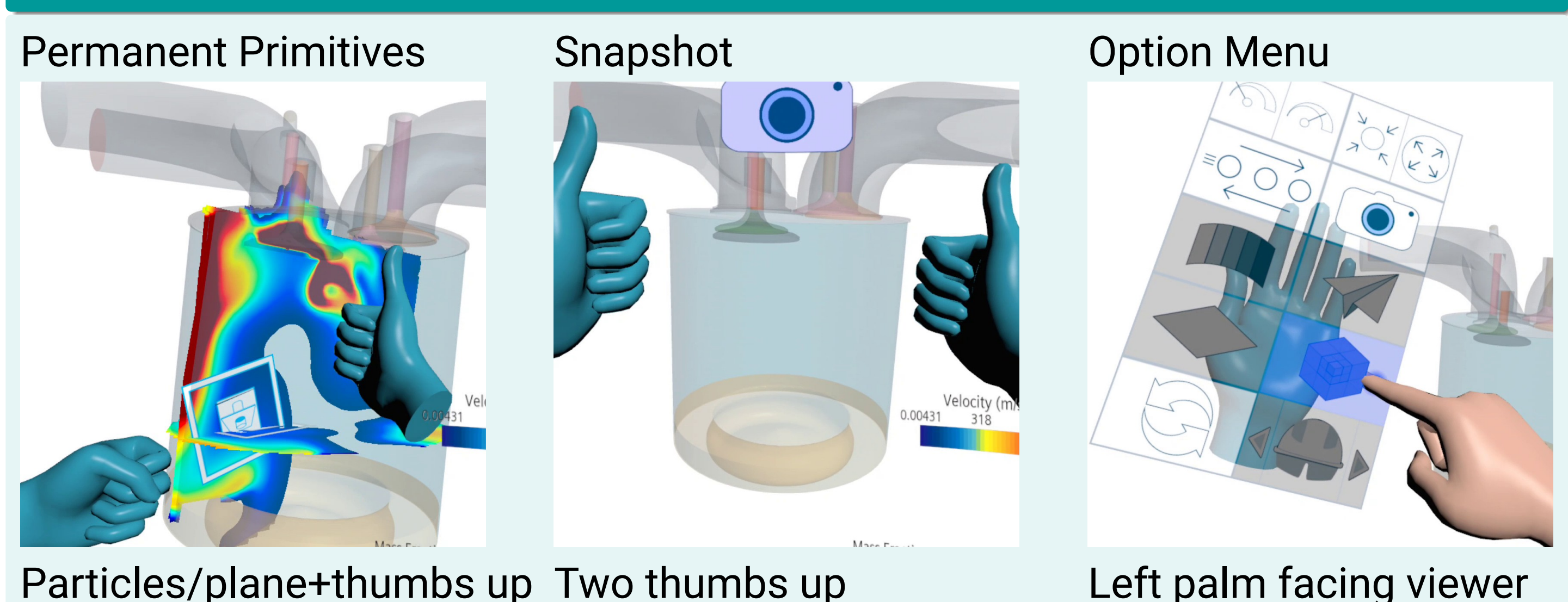
Visualization Gestures



Navigation Gestures



Special Gestures



Note that in these images, the pass-through environment has been removed for clarity, and accepted gestures are indicated by the hand model color being changed to cyan.

Conclusions

Our preliminary user evaluations indicate that hand-gesture-based AR indeed offer a more natural and less intrusive experience than controller-based VR, with an easier learning process. The main bottleneck of our viewer is the rendering performance of mobile hardware like the Meta Quest 3, which restricts usage to datasets of low to medium complexity.

Future Work

- A formal quantitative study to evaluate the user experience
- Since WebGL requires careful optimizations, identifying bottlenecks to improve rendering performance is the top priority
- A digital twin with the goal of registering virtual geometry with the corresponding real world objects, computing the simulation data, and then exploring the simulation data in the real world

- [1] Siemens Digital Industries Software. *Simcenter STAR-CCM+*. <https://www.plm.automation.siemens.com/global/en/products/simcenter/STAR-CCM.html>. 2025.
- [2] M. B. Pereira, F. d. S. Lancelotte, T. M. de Classe, and A. C. B. Garcia. "Simulation Sickness in Virtual Reality Games, How to Relieve it - Systematic Literature Study". In: *Proc. of the 26th Symposium on Virtual and Augmented Reality. SVR '24*. Manaus, Brazil: Association for Computing Machinery, 2024, pp. 168–176.
- [3] H. L. Ho, Y. Wang, A. Wang, L. Bai, and H. Ren. "Web-based Augmented Reality with Auto-Scaling and Real-Time Head Tracking towards Markerless Neurointerventional Preoperative Planning and Training of Head-mounted Robotic Needle Insertion". In: *IEEE International Conf. on Robotics and Biomimetics*. 2024.